

I recommend that all students work on **Model 1: Environmental State Classifier** because:

- It uses simple sensor data (temperature, humidity, pressure, proximity).
- Data collection is manageable within the remaining time.
- It covers all essential TinyML concepts.
- Students can still demonstrate deployment on the Arduino Nano 33 BLE Sense.
- It avoids the complexity of CNNs, audio processing, and anomaly detection.

Revised Presentation Template

SentinelAI Mini Project

TinyML Environmental State Classifier

Presentation Duration: 10–12 minutes

Slides: 12–15 slides

Slide 1 – Title Slide

Project Title

Example:

SentinelAI: TinyML-Based Environmental State Classification on Arduino Nano 33 BLE Sense

Include:

- Student Name
- Index Number
- Course
- Lecturer
- Date

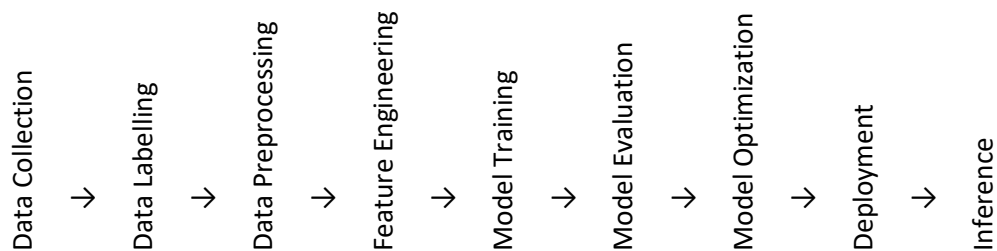
Slide 2 – Problem Statement

Describe:

- What problem is being solved?
- Why environmental monitoring matters?
- Why TinyML instead of cloud AI?

Slide 3 – Project Overview

Show the complete pipeline.



Slide 4 – Hardware and Software Platform

Hardware

- Arduino Nano 33 BLE Sense
- Built-in sensors used

Software

- Arduino IDE
- Edge Impulse
- TensorFlow Lite for Microcontrollers

Include a photo of the board.

Slide 5 – Data Collection

Explain:

Sensors Used

Sensor	Variable
HTS221	Temperature
HTS221	Humidity
LPS22HB	Pressure
APDS9960	Proximity

Dataset

- Number of samples collected
- Sampling frequency
- Environmental classes

Example:

- Normal
- Hot & Dry
- Cold & Humid
- Polluted

Include screenshots of serial monitor or Edge Impulse data acquisition.

Slide 6 – Data Exploration and Labelling

Present:

Dataset Statistics

Class	Samples
Normal	
Hot & Dry	
Cold & Humid	
Polluted	

Slide 7 – Data Preprocessing and Feature Engineering

Explain:

Features Used

Raw Features

- Temperature
- Humidity
- Pressure
- Proximity

Optional Derived Features

- Heat Index
- Temperature/Humidity Ratio

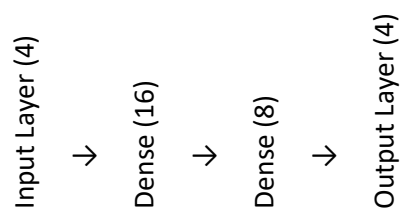
Discuss:

- Missing values
- Outliers
- Normalization

Slide 8 – Model Development

Neural Network Architecture

Example:



Include:

- Activation functions
- Epochs
- Batch size
- Learning rate

Slide 9 – Model Training Results

Present:

Training Curves

- Training Accuracy
- Validation Accuracy
- Training Loss
- Validation Loss

Discuss:

- Overfitting
- Underfitting

Slide 10 – Model Evaluation

Present:

Confusion Matrix

Performance Metrics

Metric	Value
Accuracy	
Precision	
Recall	
F1-score	

Slide 11 – Model Optimization

Explain:

Quantization

Compare:

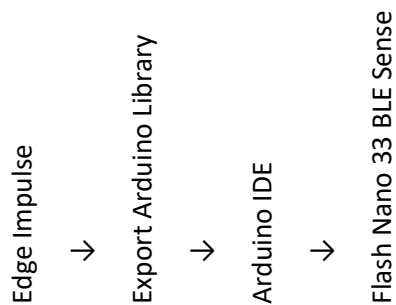
Metric	Float32	Int8
Model Size		
RAM Usage		
Latency		

Discuss why quantization is important for TinyML.

Slide 12 – Deployment to Microcontroller

Show:

Deployment Workflow



Include screenshots of:

- Edge Impulse Deployment page
- Arduino IDE

Slide 13 – Inference and Live Testing

Present:

Real-Time Inference

Example output:

Prediction: Normal

Confidence: 0.92

Latency: 5 ms

Discuss:

- Inference speed
- Memory usage
- Prediction confidence

Include photos or screenshots.

Slide 14 – Challenges and Lessons Learned

Discuss:

- Data collection challenges
- Sensor noise
- Model accuracy issues
- Deployment difficulties

Slide 15 – Conclusion and Future Work

Key Findings

- Model achieved XX% accuracy.
- Successfully deployed on Arduino Nano 33 BLE Sense.
- Real-time inference demonstrated.

Future Improvements

- Larger dataset
- Additional sensors
- More environmental classes
- BLE connectivity

1. Short Demonstration Video (2–3 minutes)
2. Final Report (3–5 pages)

Simplified Marking Rubric

Component	Weight
Data Collection & Labelling	20%
Data Analysis & Feature Engineering	15%
Model Design & Training	20%
Model Evaluation	15%
Deployment & Inference	20%
Presentation & Q&A	10%
Total	100%